Yan-Shuo Pan

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Summary

- Data scientist and AI engineer with a Ph.D. in Statistics, delivering production ML and AI agent systems for semiconductor and textile manufacturing clients.
- Builds end-to-end pipelines from raw sensor/trace data to deployed models: feature engineering, tree-based and deep learning models, time-series forecasting, and automated reporting for non-technical stakeholders.
- Develops LLM applications with RAG retrieval (embeddings, hybrid search) and multi-step agent workflows; peer-reviewed researcher in high-dimensional variable selection.

Technical Skills

Languages: Python, R, SQL

ML / DL: scikit-learn, XGBoost, LightGBM, PyTorch, TensorFlow

Infrastructure & Tools: Git, Docker, FastAPI, PostgreSQL, GitHub Actions (CI/CD)

LLM & Agents: OpenAI API, Claude API, RAG pipelines (embeddings, hybrid search), AI agent workflow design

Specialties: High-dimensional variable selection, causal inference, time-series forecasting, anomaly detection, signal processing

Work Experience

Morale AI Jul 2025 – Present
Data Scientist / AI Agent Developer (Part-time)

- Building AI agent systems that automate industrial data analysis — ingestion, anomaly detection, and report generation — for manufacturing clients without in-house data teams.
- Trueway (Textile): Built an AI agent that recommends knitting machines and yarn suppliers for a target fabric type and dye class, integrating 100K+ production records from 7 ERP sources; backtested recommendations against historical defect rates, replacing experience-based manual machine assignment.
- Yu Yuan Textile: Built per-fabric CatBoost models on three years of dyeing and finishing production data, predicting batch-level loss rate within 2–3 percentage points (test MAE); used SHAP analysis to pinpoint loss-driving process stations, powering an AI diagnostic agent for yield improvement.

National Tsing Hua University Jul 2025 – Present
Postdoctoral Researcher, Institute of Statistics

- Developing MPCGA, a multi-path tree-based variable selection algorithm that outperforms Lasso by up to 14 points in test accuracy; applying it to exhaled-breath VOC lung-cancer detection, identifying a compact panel of about 12 biomarkers for low-cost sensor design.
- Developing a multiply robust average-treatment-effect estimator for causal inference under high-dimensional confounders (first-author manuscript in preparation).

Compulsory Military Service, Taiwan Feb 2025 – Jun 2025

National Tsing Hua University — Industry Collaborations 2020 – 2022
Research Collaborator (Micron, Etron)

- Micron (2020): Predicted wafer etch depth from per-second tool trace signals (104 sensor variables per wafer); high-dimensional greedy variable selection reached a median test R^2 of 0.6 over 1,000 resampling splits, outperforming XGBoost and deep-learning baselines that degenerated to mean prediction.
- Etron (2022): Forecast monthly shipment volumes for 9 memory product lines from 6+ years of demand history; stabilized volatile small-sample forecasts with adaptive-Lasso and bootstrap stacking ensembles.

Projects

Stat-Assistant — LLM Knowledge Platform for Statistical Research 2025 – Present
github.com/YanShuoPan/stat-assistant

- Built a full-stack RAG platform (FastAPI, Next.js, PostgreSQL) where researchers upload papers that are parsed into a retrieval-ready knowledge base, and chat with an LLM assistant grounded in hybrid search (semantic embeddings + keyword) spanning 19 statistical domains.

Qtrading — Automated Taiwan Stock Screener 2025
github.com/YanShuoPan/Qtrading

- Fully automated daily stock screening pipeline with technical-indicator analysis, K-line chart generation, and multi-user LINE notification delivery, scheduled via GitHub Actions CI/CD.

Education

National Tsing Hua University Sep 2018 – Oct 2024
Ph.D. in Statistics

National Central University 2015 – 2018
B.S. in Mathematics | Minor in Computer Science

Publications & Research

On Lasso with Many Time Series Predictors and Moderately Dense Coefficients *2026*
C.-Y. Sin, **Y.-S. Pan**, C.-K. Ing | *Japanese Journal of Statistics and Data Science* | doi:10.1007/s42081-026-00347-z
Unifies Lasso theory for time-series predictors under hard and soft sparsity, showing that substantially fewer candidate variables suffice under moderately dense coefficients.

MPCGA: A Tree-Based Chebyshev Greedy Algorithm *First author, under review*
Generalizes greedy selection to multi-path tree search, achieving more stable and accurate variable selection than classical penalized methods in high-dimensional logistic regression.

Multiply Robust ATE Estimator with High-Dimensional Covariates *First author, in preparation*
Proposes a multi-step Chebyshev-greedy-based doubly robust estimator for average treatment effect estimation under high-dimensional confounders.

Honors & Awards

Ching-Zong Wei Memorial Award — Chinese Institute of Probability and Statistics (2023)

Cultivate Outstanding Doctoral Students Scholarship — NSTC (2019–2023)

Presidential Scholarship — National Tsing Hua University (2019–2023)